AMARNATH S. PATEL

apatel6ty@protonmail.com | apatel.co | github.com/jeebuscrossaint

EDUCATION

University of Central Florida — **Undergraduate Student** **4.00 GPA**
Physics (Optics & Lasers), Mathematics, Computer Science Minor — August 2025 - Present

- Relevant Coursework: Geometric Optics, Matrix & Linear Algebra, Quantum Information Processing, Discrete Computational Structures, Theoretical Methods for Physics

Florida Atlantic University — **3.66 GPA**
Computer Science coursework - High School Diploma (111 Credit Hours) — August 2021 - May 2025

- Relevant Coursework: Deep Learning, Data Structures & Algorithms, Computer Logic Design, Structured Computer Architecture

PROFESSIONAL EXPERIENCE

Instrumentation Software Engineer — Research — August 2025 - Present
UCF Astrophotonics Lab, CREOL — Dr. Stephen Eikenberry

- Built an automated digital holography pipeline coordinating 4 lab instruments (HP 8168E tunable IR laser, DiCon GP700 7-port fiber switch, Thorlabs MPC320 polarization controller, Xenics Bobcat 320 GigE InGaAs camera) over GPIB/VISA, RS-232, and GigE Vision across port × wavelength C-band sweeps (1525–1575 nm).
- Developing PolyOculus, control software for an 8-telescope photometric observation array with INDI-protocol mount control, focuser automation, and RA/Dec coordinate slewing.

Software Engineer — Research — March 2026 - Present
UCF Physics Department — Dr. Zhongzhou Chen

- Developing ESTELA, an automated exam generation system that produces multi-version isomorphic physics exams from a 615-problem bank, supporting scalable and equitable assessment infrastructure for introductory STEM courses.
- Funded by NSF award 2421299 and Gates Foundation INV-076932.

Teaching Assistant — August 2024 - May 2025
Florida Atlantic University

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, and grading. Part-time (10h/week).

Software Engineer — November 2023 - May 2025
Advanced Experimental Vehicles, Florida Atlantic University

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
- Won 2nd Place in Division and Lockheed Martin Award for "Highest Level of Engineering Excellence."

AI/ML Software Engineer — Research — January 2024 - March 2025
FAU Grant-Funded AI Safety Research Project, Florida Atlantic University

- Benchmarked 10 coherence and detection methods (GPT-2 perplexity, BERT NSP, RoBERTa, LSA, NLI, burstiness) on a 255K-passage corpus to evaluate identification of AI-generated text.
- Identified GPT-2 perplexity as the strongest discriminator (3.35× separation, 17.5 vs 58.5); BERT NSP failed to distinguish coherent from incoherent text.
- Built 4 adversarial evasion pipelines (iterative rewriting, tree-search decoding, list-branching, RL) generating 3,125 sequences; characterized a consistent quality–evasion tradeoff. Presented at Wilkes Honors College Symposium.

PROJECTS

ESTELA - Problem Bank Visualizer & Exam Generator — March 2026 - Present

- Built in Rust (Tauri 2) with a vanilla JS frontend; parses 615 problems across 29 YAML banks spanning 13 topic areas and 11 question types.
- Generates up to 10 isomorphic exam versions with auto-generated answer keys; renders LaTeX math via KaTeX and exports to 4 formats (LaTeX, Word, HTML, ZIP bundle).

Photonic Lantern Digital Holography Automation — January 2026 - Present

- Automated data acquisition and processing pipeline for photonic lantern characterization via off-axis digital holography across port × wavelength C-band sweeps (1525–1575 nm).
- Implements FFT sideband demodulation, order-4 Butterworth filtering, quadratic-phase correction, and 8-mode LP decomposition at ≈98% reconstruction fidelity.

PolyOculus — January 2026 - Present

- INDI-protocol mount control, automated focuser, RA/Dec slewing, and backlash compensation for an 8-telescope photometric observation array; part of ongoing astrophotonics research at UCF CREOL.

CELERIS — Metalens Design Pipeline & From-Scratch RCWA Solver — 2026

- From-scratch, validated electromagnetic solver for metalens design via rigorous coupled-wave analysis (1D TE/TM + 2D vectorial, scattering-matrix recursion, Sellmeier/tabulated dispersion); cross-validated against the grcwa and Stanford S^4 solvers (agreement to 1e-7) and reproduces published metalenses (Khorasaninejad 2016, Chen 2018).
- End-to-end design→analysis→GDSII pipeline: meta-atom library sweeps, achromatic & Pancharatnam–Berry design, optical-analysis battery, GPU-accelerated far-field propagation, and fabrication-ready layout export. C++/CUDA with Python bindings; manuscript in preparation.

SKILLS

Languages: C/C++, CUDA, Rust, Python, VHDL, Shell (Fish, Bash, tcsh)

Tools & Frameworks: Docker, Git, CMake, XMake, NumPy, SciPy, INDI, VirtualBox, VMware, QEMU
Operating Systems: Linux Distributions, BSD, Windows